AP PRECALCULUS

Unit 1 • Worksheet 1.5E • Answer key

Worksheet 1.5E • Polynomial Functions and Complex Zeros

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

Which function is even?

(A) $x^3 - x$

(B) $x^4 + 3x^2 - 1$

(C) $x^2 + x$

(D) $(x-1)^2$

Solution

Answer: (B). Only even powers.

Question 2

NO CALCULATOR

Which function is odd?

(A) $x^4 + x$

(B) $x^3 + 1$

(C) $x^3 - 5x$

(D) $|x|$

Solution

Answer: (C). $f(-x) = -f(x)$.

Question 3

NO CALCULATOR

The graph of an odd function passes through $(3, -7)$. Which point must also be on the graph?

(A) $(-3, -7)$

(B) $(3, 7)$

(C) $(-3, 7)$

(D) $(7, -3)$

Solution

Answer: (C). Symmetric about the origin: $f(-3) = -f(3) = 7$.

Question 4 (Free response) NO CALCULATOR

Let $p(x) = (x-1)^2(x+3)(x^2+1)$.

(a) Find the degree of p and all of its complex zeros with their multiplicities.
(b) Describe the graph of p at each x-intercept.
(c) Is p even, odd, or neither? Justify.

Solution
(a) Degree 5. Zeros 1 (2), -3 (1), $\pm i$ (1 each).
(b) Touches at 1, crosses at -3.
(c) $p(1) = 0$ but $p(-1) = (4)(2)(2) = 16$: $p(-1) \neq p(1)$ and $p(-1) \neq -p(1)$. Neither.